

HRIDIK HINGORANI

Curriculum Vitae

Champaign, IL

• * (447) 902-5994

• hridikh2@illinois.edu

• hridikh.github.io/portfolio

github.com/HridikH linkedin.com/in/HridikH



Education

Aug 2023 – **BS Computer Engineering, Minor in Mathematics**, *University of Illinois at Urbana-Champaign*
May 2027
(expected)

Coursework ECE 470 Robotics, ECE 486 Control Systems, ECE 484 Principles of Safe Autonomy, ECE 391 Operating Systems, ECE 494 Deep Learning for Computer Vision

Experience

May 2026 – **AI Software Engineering Intern**, *Software Associates*, Remote

- Built a local-first voice assistant on embedded Linux and Apple Silicon: Whisper STT, LLM routing through Ollama with cloud fallback, Coqui TTS, a FastAPI service layer, and an Electron client. Ran MLX LoRA fine-tuning to adapt the local model's response behavior.
- Diagnosed and fixed audio-pipeline timing failures, device interface faults, and I2C/UART peripheral errors that were dropping utterances mid-stream, taking capture from intermittent to reliable end to end.
- Shipped a Python and Flask trading service on the Interactive Brokers REST API: a five-indicator weighted signal engine (RSI, MACD, Bollinger Bands, moving averages, ATR), automated stop-loss, take-profit and trailing-stop risk logic, and a live position and P&L dashboard.

May 2025 – **AI Software Development Intern**, *HealthArk Insights*, Mumbai, India

- Built Python-based document-processing features including summarization, web scraping, and dynamic formatting tools tailored to client specifications.
- Integrated client feedback into production workflows by gathering requirements, validating features, and refining document-management functionality. Features shipped to production.

Projects

Jan 2024 – **StethoSpy: Electronic Stethoscope**, *Project Lead, I-MADE UIUC*

Aug 2025 Prototype electronic stethoscope for heart-sound acquisition and murmur screening, aimed at rural and low-infrastructure settings. Custom PCB captures heart sounds; an ADC/DAC pipeline processes the analog signal to flag cardiac irregularities for follow-up. Not clinically validated. Owned hardware design and signal processing end to end.

8-Channel EEG Acquisition Board, *Independent research*

8-channel, 2-layer EEG signal acquisition board in KiCad. Op-amp analog front end with low-pass filtering for microvolt-level biopotential acquisition; schematic capture through PCB layout designed for high signal integrity across all eight channels.

UR3 Vision-Guided Contour Drawing, *Python, ROS, OpenCV*

Closed-form analytic inverse kinematics for a 6-DOF arm, written from scratch, with product-of-exponentials forward kinematics evaluated through matrix exponentials. Drives a UR3 to draw an image: Canny edge detection and contour simplification mapped onto a 20 cm square drawing area, published as joint targets over ROS and held to 0.0005 rad per joint. A Pikachu test image came out as 142 strokes, drawn in 7 minutes.

Real-Time Task-Space Impedance Control: CRS Arm, C, TMS320F28335 DSP

Task-space PD impedance control on a three-link arm, mapped to joints through the Jacobian transpose and closed at 1 kHz on a timer interrupt. Reads joint encoder angles and outputs joint torque commands directly to the amplifiers; it does not send position setpoints to a servo controller. Per-waypoint stiffness modes give selective compliance, with per-joint Coulomb and viscous friction compensation and torque saturation at 5 N-m per joint. Contact tasks use a feedforward downward force commanded through the motor torque constant, with no force sensor and no closed-loop force feedback, validated afterward against a physical scale. Written in C on a TI TMS320F28335. Peg insertion, zig-zag tracking, and pushing an egg without breaking it.

We Are Kernel: RISC-V OS, C, RISC-V

A RISC-V kernel from scratch, roughly 6,000 lines of C: Sv39 paging, VirtIO, a FAT filesystem, fork/exec/exit, pipes, a shell with redirection, and preemptive multitasking.

Animatronic Grogg, Python, C++, KiCad, ROS

ECE 445 senior design capstone, currently in planning. A 22-DOF bipedal animatronic. Planned scope covers the software, the LLM voice pipeline (local LLM with Claude API, Whisper speech-to-text, Coqui text-to-speech), and the mechanical design, on a custom PCB for power.

Jupiter AI Voice Assistant, Python, MLX, Qwen3, Whisper

Fully local voice assistant for Apple Silicon, no cloud and no API keys. Whisper STT, a Qwen3 router and main model on MLX with speculative decoding and a persistent prompt cache, and Kokoro streaming TTS. Continuous VAD gives barge-in that halts generation mid-sentence. 0.3 to 0.6 s first token, warm.

Inverted Pendulum: Pole Placement and Observer, MATLAB, Embedded

Pole-placement state feedback with a Luenberger observer, built as two decoupled 2-state blocks and gain-scheduled across the upright and hanging linearizations. Friction parameters identified from logged step-response data.

Teaching Experience

- Jan 2026 – present **Course Assistant, ECE 385 (FPGA Design), University of Illinois at Urbana-Champaign**
Leads lab sections for 60+ students on SystemVerilog, FPGA timing closure, and hardware-software co-design. Debugs student implementations across RTL logic, timing constraints, and board-level signal integrity.
- Jan 2025 – Dec 2025 **Course Assistant, MATH 257 (Linear Algebra), University of Illinois at Urbana-Champaign**
Taught Python-based linear algebra lab sections to approximately 200 students, supporting hands-on computational exercises.

Research

- Ongoing **Project Lead, Brainwaves Team, NeuroTechX UIUC**
Leads the Brainwaves project team, overseeing the club's podcast and social media output. EEG signal-processing and neural-network research through the club.
- Ongoing **Independent EEG Research, Self-directed**
Independently designed and built the 8-channel EEG acquisition board listed under Projects; separate from the club work.

Leadership

- Jan 2025 – present **Event Lead, then Senior Coordinator, National Team, Hindu YUVA**
Manages the UIUC chapter and the event team. Organizes large-scale Hindu cultural and community events at both chapter and national scope.
- Jan 2024 – Aug 2025 **Project Lead, StethoSpy, I-MADE UIUC**
Led the StethoSpy project (see Projects), owning hardware design and signal processing end to end.

Skills

- Robotics & controls Closed-form and analytic IK, product of exponentials, impedance and compliance control, LQR, PID, RK4 integration, motion planning (A*, RRT*), ZMP gait planning, ROS, URDF
- Languages C/C++, Python, SystemVerilog, RISC-V Assembly, MATLAB, Rust (application-level)

Embedded & hardware Real-time Linux, bare-metal firmware, preemptive scheduling, I2C/UART/ADC/GPIO/PWM, SPI (loopback), ESP32, Raspberry Pi, FPGA, KiCad, Git

Off the Clock

Writing Poetry and literary fiction. The poems lean on Mumbai-specific references, Hinglish code-switching, and language kept precise. Posted, fairly regularly, on Instagram.